



MALAWI UNIVERSITY OF SCIENCE AND TECHNOLOGY

MALAWI INSTITUTE OF TECHNOLOGY

APPLIED STUDIES DEPARTMENT

2021/22 END OF SEMESTER I EXAMINATIONS

PHYS-111: MECHANICS AND THERMAL PROPERTIES OF MATTER

[100 marks]

Examiners: R. Chimombo, L. Chisanu, C. Daka & O. Kachaje.

DATE: **TIME:** 3hrs:(9:00am - 12:00pm)

INSTRUCTIONS

1. Write your **Registration Number** on each page of your answer sheets.
2. Start each question on a fresh page.
3. This paper contains 5 **pages** and 5 **QUESTIONS**, please check.
4. Answer **ALL** questions.
5. Show your working clearly.
6. Marks allocated for each question are indicated against it.
7. Round off your final answer to 2 decimal places.

LIST OF PHYSICAL CONSTANTS

Acceleration due to gravity	g	9.8 m/s^2
Universal gravitation constant	G	$6.67 \times 10^{-11} \text{ N.m}^2/\text{kg}^2$
Latent heat of fusion of water	L_f	$3.34 \times 10^5 \text{ J/kg}$
Latent heat of vaporisation of water	L_v	$22.6 \times 10^5 \text{ J/kg}$
Coefficient of linear expansion for aluminium	α	$2.4 \times 10^{-5} / ^\circ\text{C}$

Question 1.

- A. Calculate the volume in SI unit of a wire of length 125 cm and diameter 0.65 mm? **[4 marks]**
- B. Check the correctness of the physical equation $v^2 = u^2 + 2as^2$. **[6 marks]**
- C. A car travels 100 km to the east. If the first half of the distance is driven at 50 km/h and the second half at 100 km/h. What is its average velocity in SI units? **[6 marks]**
- D. A cricketer can throw a ball a maximum horizontal distance of 100 m. If the ball is thrown with the same effort vertically, what is the maximum height that the ball can attain? **[4 marks]**

Question 2.

- A. A car moves with a velocity of 30 m/s due north 60° west. Calculate the components of velocity due north and west. **[4 Marks]**
- B. A plane is moving horizontally through the sky at a constant speed of 250 m/s. A package is dropped from the plane and travels a horizontal distance 1 km from the point of release where it lands on the target, if air resistance is negligible, find the height from which the package was released. **[6 marks]**

- C. Two packing crates of masses 10 kg and 5 kg are connected by a light string that passes over a frictionless pulley as in Figure 2. The 5 kg crate lies on an incline plane surface of angle 40° . If the coefficient of kinetic friction between the crate and the plane surface is 0.21. Determine:

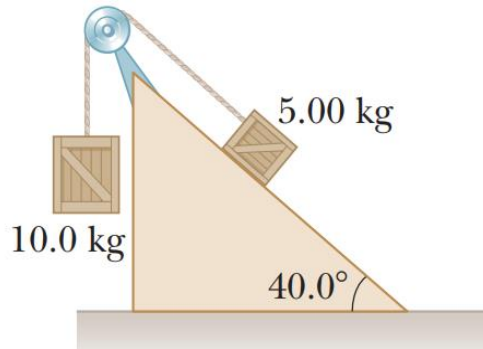


Figure 1

- i. Free-body diagrams for 10 kg and 5 kg crates. [3 marks]
- ii. Magnitude of acceleration of the system [7 marks]

Question 3

- A. A 1130 kg car is being pulled up a frictionless ramp at a constant speed, as shown in **Figure 2**. The cable makes an angle of 31° above the surface of the ramp, and the ramp itself rises at 25° above the horizontal. The car moves up 20 m parallel to the incline. What will be the work done by the gravity? [4 marks]

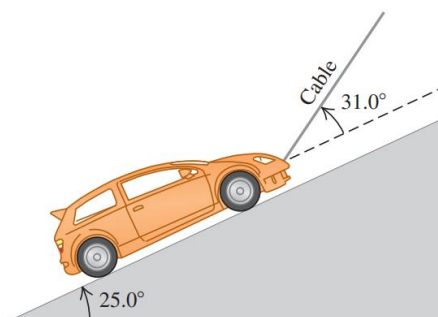


Figure 2.

- B. A rough slope is inclined at $\tan^{-1}\left(\frac{3}{4}\right)$ to the horizontal. A box of mass 25 kg slides from the rest a distance of 10 m down the line of greatest slope with a speed of 14 m/s. Find the work done by friction [6 marks]
- C. Explain how collision is applied in the automobile industry. [4 marks]
- D. A bullet which has velocity 150 m/s and mass 200g sticks to the stationary 4-kg block as in **figure 3**. They move together after the collision. What is the height the system has after the collision? [6 marks]

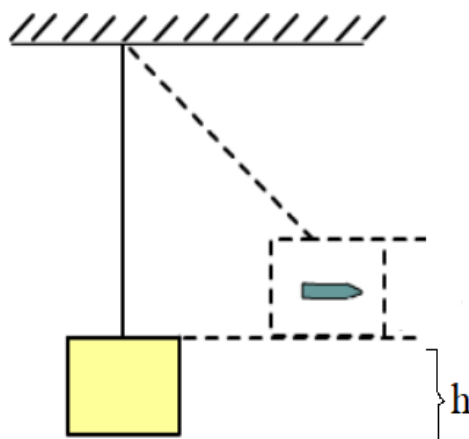


Figure 3.

Question 4

- A. The maximum tension that a 0.50 m string can tolerate is 14 N. A 0.25 kg ball attached to this string is whirled in a vertical circle. What is the maximum speed of the ball at the top of the circle? [5 marks]
- B. A minute hand of a clock is 5 cm long . Calculate the linear speed of an ant sitting at its tip [5 marks]

Question 5

- A. Define heat capacity [2 marks]
- B. The triple points of neon and carbon dioxide are 24.57 K and 216.55 K respectively. Express these temperatures on the Celsius and Fahrenheit scales. [6 marks]
- C. A 10 kW drilling machine is used to drill a bore in a small aluminium block of mass 8.0 kg. What is the rise in temperature of the block in 2.5 minutes, assuming 50% of power is used up in heating the machine itself or lost to the surroundings? Specific heat of aluminium is 0.91 J/g.K. [5 marks]
- D. Calculate the heat required to
- i. Change 2.0 kg of ice at 0°C to water at 0°C [2 marks]
 - ii. Change 0.6 kg of water at 100°C to steam at 100°C [2 marks]
- E. Show that the coefficient of area expansion (γ) is twice the coefficient of linear expansion α . that is, $\gamma = 2\alpha$. [8 marks]
- F. An aluminium sphere has a radius of $r=3.00$ mm at 100°C. What is its volume at 0°C? [5 marks]

END OF EXAMINATION PAPER